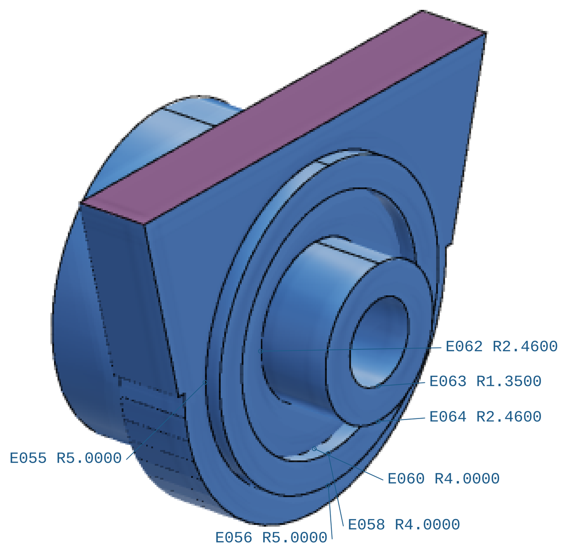
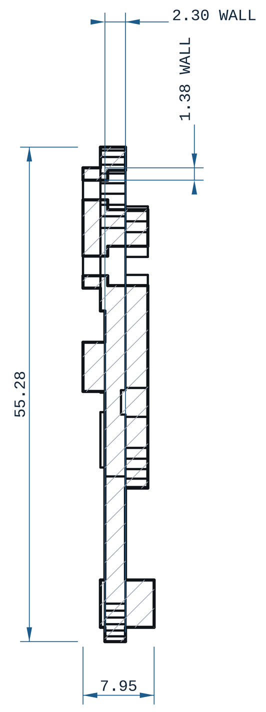
MICRODUCK-SHIN — NOMINAL mm

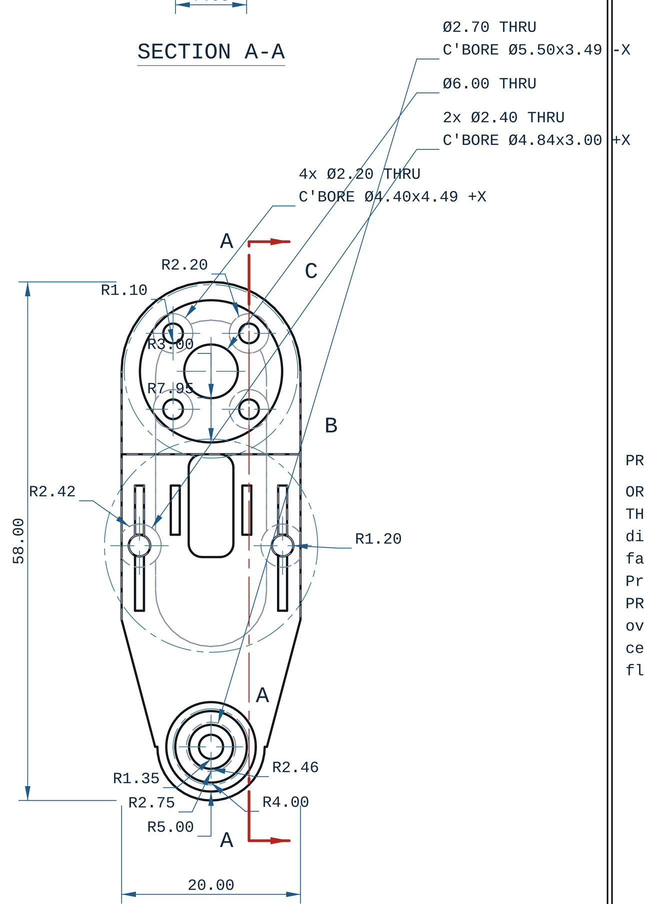
outline envelope length	7.9500
outline envelope width	20.0000
outline envelope height	58.0000
hole 1 diameter (along x)	Ø6.0000
hole 1 depth	7.2500
hole 1 position from the Y datum face	10.0000
hole 1 position from the Z datum face	48.0000
counterbore 2 diameter (along x)	Ø5.5000
counterbore 2 depth	3.4900
counterbore 2 position from the Y datum face	10.0000
counterbore 2 position from the Z datum face	6.0000
counterbore 3 diameter (along x)	Ø4.8400
counterbore 3 depth	3.0000
counterbore 3 position from the Y datum face	2.0000
counterbore 3 position from the Z datum face	28.5000
counterbore 4 diameter (along x)	Ø4.8400
counterbore 4 depth	3.0000
counterbore 4 position from the Y datum face	18.0000
counterbore 4 position from the Z datum face	28.5000

CANNOT DETERMINE — chamfer: A4 requires size and angle on every chamfer; cecad.inspect has no chamfer classifier — a chamfer is a planar face bridging two faces at an angle and nothing here identifies one SETTLED BY: a chamfer classifier in cecad.inspect (planar face whose two neighbours meet it at equal angles), or the builder declaring its chamfers in the part record





SECTION A-A



RIGHT VIEW

-Y > Z ^

MICRODUCK SHIN

2:1
A0 / THIRD ANGLE / mm
7.95 x 20.00 x 58.00
3.39 cm3
CAD SOURCE: part:microduck-shin
DATUM: XYZ MINIMUM ENCLOSING PLANES
XYZ origin mm: 32.5000, 12.0000, -12.2230
GENERAL TOLERANCE CANNOT DETERMINE — SEE PRINT / DFM NOTES UNLESS STATED.
DRAFT — NOT RELEASED

FRAME OCCUPANCY 99.88%

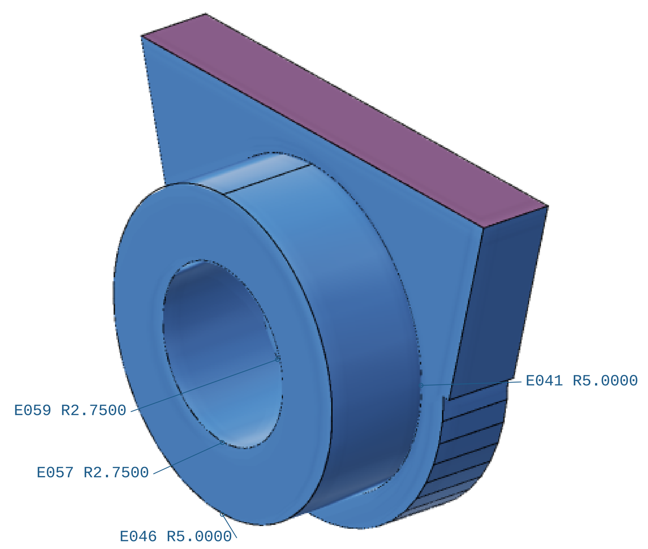
RENDER 1 / SCALE 7.94093:1

ISOMETRIC 1 FRONT LEFT / DETAIL A

MEASURED FEATURES — mm

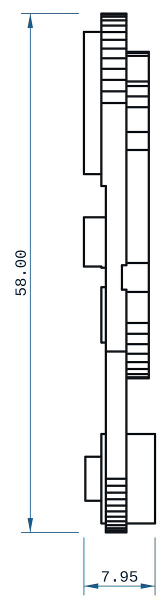
counterbore 5 diameter (along x)	Ø4.4000
counterbore 5 depth	4.4900
counterbore 5 position from the Y datum face	5.7570
counterbore 5 position from the Z datum face	43.7570
counterbore 6 diameter (along x)	Ø4.4000
counterbore 6 depth	4.4900
counterbore 6 position from the Y datum face	5.7570
counterbore 6 position from the Z datum face	52.2430
counterbore 7 diameter (along x)	Ø4.4000
counterbore 7 depth	4.4900
counterbore 7 position from the Y datum face	14.2430
counterbore 7 position from the Z datum face	43.7570
counterbore 8 diameter (along x)	Ø4.4000
counterbore 8 depth	4.4900
counterbore 8 position from the Y datum face	14.2430
counterbore 8 position from the Z datum face	52.2430
hole 9 diameter (along x)	Ø2.7000
hole 9 depth	4.3000
hole 9 position from the Y datum face	10.0000

CANNOT DETERMINE — rib: A4 requires thickness, height and pitch on every rib; nothing here groups faces into a rib SETTLED BY: a rib classifier (parallel thin prisms standing off a wall), or the builder declaring its rib pitch



PRINT / DFM

ORIENTATION: upright, Z-up (8 x 20 x 58 mm)
THINNEST WALL, WHOLE SOLID (RAY MINIMUM): 0.0280 mm — a distance through material, NOT a printable wall where it falls on an edge, a fillet run-out or a chamfer tip. Printability is judged on the next line.
PRINTABLE MINIMUM WALL: 1.0000 mm (smallest median wall over a 3.1340 mm neighbourhood, cecad.inspect.printable_wall_detail) against a 0.8000 mm floor (2 perimeters @ 0.4 mm nozzle): VERDICT PASS



FRONT VIEW

-X > Z ^

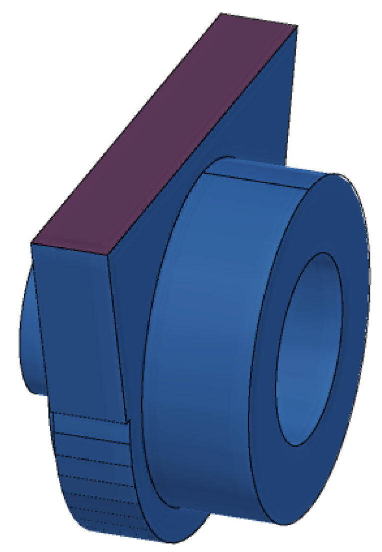
RENDER 2 / SCALE 7.94093:1

ISOMETRIC 2 FRONT RIGHT / DETAIL A

MEASURED FEATURES — mm

hole 9 position from the Z datum face	6.0000
hole 10 diameter (along x)	Ø2.4000
hole 10 depth	2.3000
hole 10 position from the Y datum face	2.0000
hole 10 position from the Z datum face	28.5000
hole 11 diameter (along x)	Ø2.4000
hole 11 depth	2.3000
hole 11 position from the Y datum face	18.0000
hole 11 position from the Z datum face	28.5000
hole 12 diameter (along x)	Ø2.2000
hole 12 depth	2.7600
hole 12 position from the Y datum face	5.7570
hole 12 position from the Z datum face	43.7570
hole 13 diameter (along x)	Ø2.2000
hole 13 depth	2.7600
hole 13 position from the Y datum face	5.7570
hole 13 position from the Z datum face	52.2430
hole 14 diameter (along x)	Ø2.2000
hole 14 depth	2.7600

CANNOT DETERMINE — draft: A4 requires every draft angle; no face here is tested for draft against a pull direction SETTLED BY: a pull direction on the part record, then the angle of every planar face to it



HOLE TABLE — 8 POSITIONS

TAG	SIZE	Y	Z	C'BORE
A1	Ø2.20 THRU	26.24	40.02	Ø4.40x4.49 +X
A2	Ø2.20 THRU	17.76	40.02	Ø4.40x4.49 +X
A3	Ø6.00 THRU	22.00	35.78	-
A4	Ø2.20 THRU	26.24	31.53	Ø4.40x4.49 +X
A5	Ø2.20 THRU	17.76	31.53	Ø4.40x4.49 +X
A6	Ø2.40 THRU	30.00	16.28	Ø4.84x3.00 +X
A7	Ø2.40 THRU	14.00	16.28	Ø4.84x3.00 +X
A8	Ø2.70 THRU	22.00	-6.22	Ø5.50x3.49 -X

every row measured off the solid; positions are model coordinates, mm. C'BORE side is the face it opens on; INT = centred in the bore, opening on neither face.

PRINT / DFM

15 horizontal bore(s) bridge at the crown: ream any bearing or press-fit bore to size
FILAMENT: PLA, ~4 g (modelled), 290 layers @ 0.2 mm

RENDER 3 / SCALE 7.94093:1

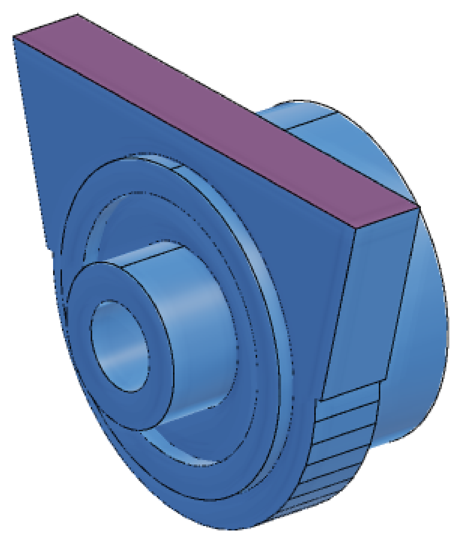
ISOMETRIC 3 REAR RIGHT / DETAIL A

MEASURED FEATURES — mm

hole 14 position from the Y datum face	14.2430
hole 14 position from the Z datum face	43.7570
hole 15 diameter (along x)	Ø2.2000
hole 15 depth	2.7600
hole 15 position from the Y datum face	14.2430
hole 15 position from the Z datum face	52.2430
groove 1 outer Ø	Ø8.0000
groove 1 core Ø	Ø4.9200
groove 1 depth	0.5000
boss / outside diameter 1	Ø15.9000
boss 1 length along its axis	1.9500
boss / outside diameter 2	Ø10.0000
boss 2 length along its axis	3.2000
boss / outside diameter 3	Ø10.0000
boss 3 length along its axis	0.5000
boss / outside diameter 4	Ø4.9200
boss 4 length along its axis	2.2900
radius R1.10 (8 edges, 8 full circles)	R1.1000
radius R1.20 (6 edges, 2 full circles)	R1.2000

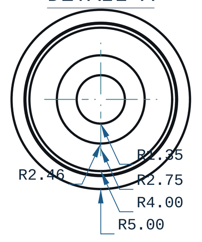
CANNOT DETERMINE — step: A4 requires every step and shoulder; nothing here classifies a shoulder between two coaxial diameters or two parallel faces SETTLED BY: a step classifier, or the section views declaring their stations

SOURCE / PROCESS
DETAIL A DISPLAY SECTION: Z = -0.2230 mm SOURCE XYZ; KEEP MIN. Raster boundaries clip the view; original geometry unchanged.
RECREATION MATERIAL: PLA
PROCESS: FDM
RECORD: ce-parts/microduck-shin/component.json
GEOMETRY SOURCE: reference/pollen-microduck-rl/assets/leg.stl (Onshape doc 804927696f06d877f3f1803e, exported by onshape-to-robot)
ORIGINAL MATERIAL / SURFACE FINISH: CANNOT DETERMINE; obtain original specification or measure original part.
PART MASS: CANNOT DETERMINE; weigh the produced part or qualify material density and actual infill before assigning mass.
FASTENER LENGTH / TORQUE / FIT CLASS: CANNOT DETERMINE; verify interface stack and test production coupon.
KNEE MATE CONTRACT: connection:spline-xl330-horn
MATING PART: part:xl330-m288-t
ANKLE MATE CONTRACT: connection:press-fit-bearing-15x10x3
MATING PART: part:bearing-15x10x3



SCALE 4:1

DETAIL A



RENDER 4 / SCALE 7.94093:1

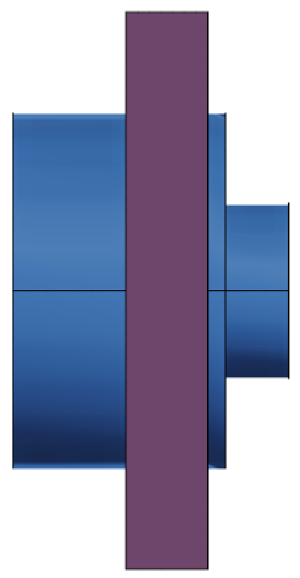
ISOMETRIC 4 REAR LEFT / DETAIL A

MEASURED FEATURES — mm

radius R1.35 (2 edges, 2 full circles)	R1.3500
radius R2.20 (12 edges, 4 full circles)	R2.2000
radius R2.42 (6 edges, 0 full circles)	R2.4200
radius R2.46 (2 edges, 2 full circles)	R2.4600
radius R2.75 (2 edges, 2 full circles)	R2.7500
radius R3.00 (2 edges, 2 full circles)	R3.0000
radius R4.00 (2 edges, 2 full circles)	R4.0000
radius R5.00 (4 edges, 4 full circles)	R5.0000
radius R7.95 (2 edges, 2 full circles)	R7.9500
thinnest wall	0.0280
slot mouth on a plane normal to x, 8 edge(s) — width	2.4000
slot mouth on a plane normal to x, 8 edge(s) — length	14.0000
slot mouth on a plane normal to x, 8 edge(s) — depth	2.4500
slot mouth on a plane normal to x, 8 edge(s) — position from the y datum	2.0000
slot mouth on a plane normal to x, 8 edge(s) — position from the z datum	28.2230
pocket mouth on a plane normal to x, 4 edge(s) — width	1.0000
pocket mouth on a plane normal to x, 4 edge(s) — length	5.5000
pocket mouth on a plane normal to x, 4 edge(s) — depth	2.4500
pocket mouth on a plane normal to x, 4 edge(s) — position from the y datum	6.0000

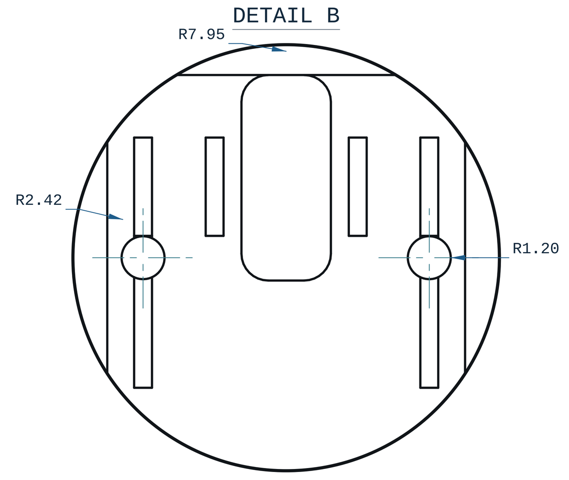
CANNOT DETERMINE — mate: A4 requires every surface that mates with another part, with the mating part named by its triad ref; the mates live in ce-assemblies/*/joints.json and are not read here SETTLED BY: reading this part's joints out of the assemblies that use it and matching each mate to a face

HIDDEN ARC LOCATORS / DATUM XYZ mm
E061 R1.3500 / SWEEP 360.0000 deg
CENTRE 3.4900, 10.0000, 6.0000
AXIS 1.0000, 0.0000, 0.0000
START 3.4900, 10.0000, 7.3500
END 3.4900, 10.0000, 7.3500



SCALE 4:1

DETAIL B



RENDER 5 / SCALE 7.94093:1

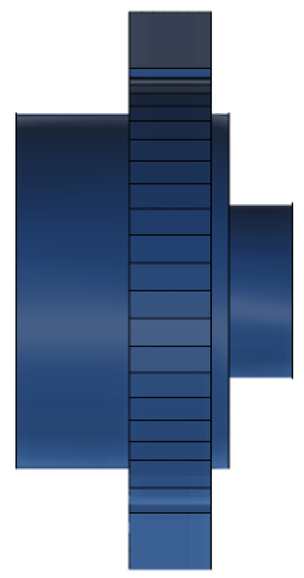
TOP-DOWN RENDER / DETAIL A

MEASURED FEATURES — mm

pocket mouth on a plane normal to x, 4 edge(s) — position from the z datum	32.4730
pocket mouth on a plane normal to x, 36 edge(s) — width	5.0000
pocket mouth on a plane normal to x, 36 edge(s) — length	11.5000
pocket mouth on a plane normal to x, 36 edge(s) — depth	5.3000
pocket mouth on a plane normal to x, 36 edge(s) — position from the y datum	10.0000
pocket mouth on a plane normal to x, 36 edge(s) — position from the z datum	32.9730
pocket mouth on a plane normal to x, 4 edge(s) — width	1.0000
pocket mouth on a plane normal to x, 4 edge(s) — length	5.5000
pocket mouth on a plane normal to x, 4 edge(s) — depth	2.4500
pocket mouth on a plane normal to x, 4 edge(s) — position from the y datum	14.0000
pocket mouth on a plane normal to x, 4 edge(s) — position from the z datum	32.4730
slot mouth on a plane normal to x, 8 edge(s) — width	2.4000
slot mouth on a plane normal to x, 8 edge(s) — length	14.0000
slot mouth on a plane normal to x, 8 edge(s) — depth	2.4500
slot mouth on a plane normal to x, 8 edge(s) — position from the y datum	18.0000
slot mouth on a plane normal to x, 8 edge(s) — position from the z datum	28.2230

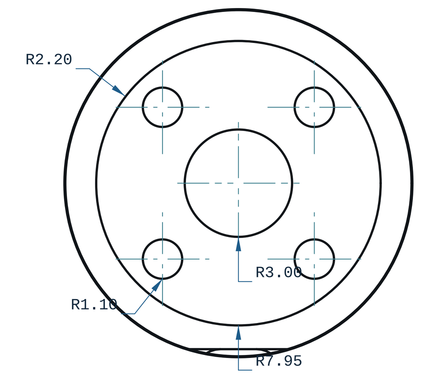
CANNOT DETERMINE — wall: A4 wants a wall thickness at every distinct station; only the thinnest wall is measurable off the solid without a station list SETTLED BY: the drawer declaring the section stations it dimensions, or a section per shell region

DRAFTING GAPS — NOT RELEASED
E009 / E047 / E061 have measured coordinate locators; section or detail leaders remain required by A.5.
Enlarged shaded feature-cluster details and non-circular feature locator audit remain incomplete.



SCALE 4:1

DETAIL C



RENDER 6 / SCALE 7.94093:1

BOTTOM-UP RENDER / DETAIL A